ADOLFO ESTEBAN ÁLVAREZ PACHECO

Senior Backend Engineer | Full Stack Developer

Node.js · TypeScript · NestJS · React · Next.js · AWS · Microservices · DDD

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SUMMARY

Computer Systems Engineer with **6+ years** of Full Stack experience, senior in **backend with Node.js, TypeScript and NestJS** and cloud architectures on **AWS** (Serverless, AWS CDK). I design **microservices**, Clean Architecture and DDD systems; junior/mid-level in frontend with React and Next.js. AI-assisted development using **SDD** and **ODD** depending on change size, plus in-house tooling to cut token usage. Experience in telecommunications, hospitality, logistics, education and HR.

SKILLS

Languages: TypeScript, JavaScript, Node.js, Python, Java. Learning: Go

Backend: NestJS, Express, HapiJS, Restify, REST, GraphQL, WebSockets, JWT, OAuth, OpenAPI

Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS, Zustand (junior/mid level)

Architecture: Microservices, Clean Architecture, DDD, SOLID, Serverless, event-driven architecture, CQRS

Databases: PostgreSQL, PostGIS, MySQL, MongoDB, DynamoDB, Redis. ORMs: Prisma, TypeORM, Drizzle, Sequelize, Mongoose

Cloud (AWS): Lambda, API Gateway, S3, SNS, Step Functions, CloudWatch, EC2, VPC, RDS, IAM, AWS CDK. Learning: Terraform, Kubernetes

Messaging: Kafka, RabbitMQ

DevOps / CI-CD: Docker, Docker Compose, Bitbucket Pipelines, GitHub Actions, GitLab, Git/Gitflow, Grafana, CloudWatch

Testing: Jest, Vitest, Supertest, Mocha

Applied AI: Claude Code, OpenCode, GitHub Copilot, n8n, SDD, ODD. Learning: RAG, LangChain, embeddings, vector databases, LLM workflows.

Practices: Code reviews, technical mentoring, Scrum/Agile, estimation, performance optimization

EXPERIENCE

INFOTEC – Senior Backend Engineer May 2023 – Mar 2026

National Internet para Todos platform (CFE Telecomunicaciones) managing the logistics operation of connectivity infrastructure across Mexico: suppliers, contracts, materials, installation sites, photo evidence and operational tracking.

- Sole backend developer on a team of 3; in the last year, became the person responsible for the projects, repositories and coordination with analysis and DevOps.
- Designed the backend architecture with Clean Architecture and **10+ microservices** (authentication, users, logistics, installations, among others) and built virtually all of the backend with NestJS.
- Platform with **118,174+ internet points** registered and operating (4G, satellite, fiber and copper), published on two public maps using geometric data types: mapa.internetparatodos.cfe.mx and cabinas.internetparatodos.cfe.mx.
- Designed REST APIs consumed by web and mobile apps (administrative staff and installers).
- Implemented file uploads, Excel/PDF reports and email notifications.
- Contributed to the frontend with React and Next.js (administrative modules).
- Defined development standards, ran code reviews and technical mentoring, and took part in estimation and architecture decisions.

Technologies: Node.js, TypeScript, NestJS, PostgreSQL, PostGIS, React, Next.js, AWS, Docker, JWT, Microservices

HAPPYGUEST – Backend & Cloud Engineer Feb 2022 – Jan 2023

Multitenant PMS for 4- and 5-star hotels (administration, restaurants, events and amenities) serving **50+ hotels** in Mexico.

- Migrated **80+ Lambda functions** from JavaScript to TypeScript within the first two weeks.
- Migrated infrastructure from Serverless Framework to **AWS CDK**: learned the tool in the first week and owned the setup of everything already in production, as well as deployments.
- Built and deployed **100+ Lambdas**; introduced Lambda Layers to remove duplicated dependencies and speed up deployments.
- Designed the full guest event-booking module and serverless flows (API Gateway, Lambda, DynamoDB, EventBridge, SNS) for push-notification reminders.

- Built reports integrating Zendesk, Grafana and internal services; used CloudWatch and Grafana for log analysis.
- Took part in cloud architecture decisions and code reviews with the project lead.

Technologies: AWS CDK, Lambda, API Gateway, DynamoDB, SNS, EventBridge, CloudWatch, RDS MySQL, TypeScript, Bitbucket Pipelines

DERBY TECHNOLOGIES – Backend Developer Nov 2021 – Jan 2022

REST APIs and new features for a logistics platform, under Scrum.

Technologies: Node.js, Restify, AWS, MongoDB

BELIEFF – Backend Developer Intern Mar 2020 – Oct 2021

Backend for a dynamic survey platform for recruiting (HR), under Scrum; first professional exposure to DynamoDB and AWS.

Technologies: Node.js, AWS, DynamoDB

ADC – Software Engineer & Technical Instructor 2018 – Present (intermittent)

Joined through a company-sponsored technical training program for high-performing students; later worked on real client projects and trained new developers.

- Backend for education, CRM, dashboards and enterprise platforms: REST APIs, GraphQL and WebSockets.
- Educational SaaS for course, assignment and grade management.
- Performance tuning with raw SQL instead of an ORM when the project required it.

Technologies: Node.js, TypeScript, NestJS, Express, HapiJS, GraphQL, PostgreSQL, MongoDB, AWS Lambda, S3

Independent – Personal projects and consulting 2026 – Present

Technical consulting and personal projects (see Projects section).

PROJECTS

alvaress – OpenCode agent github.com/Alvarez-P/alvaress

Built an OpenCode agent/harness that embeds a local decision model to route tasks and cut token usage (based on the Laya model from Hugging Face), with SDD/ODD flows, auditing and automatic validation. Plus a workflow for implementer agents modeled on how Aider works.

tickets-handler – In progress NestJS · DDD

Ticketing system with DDD, CQRS (PostgreSQL for writes, Elasticsearch for reads), event-driven architecture (Kafka/RabbitMQ) and AI-based ticket classification.

Kopere – In progress NestJS · Prisma · React PWA

PWA for rotating-savings (tanda) and personal-loan organizers; pnpm monorepo with a NestJS backend, frontend and shared package, tested with Vitest.

EDUCATION

Instituto Tecnológico de Mérida – B.Eng. Computer Systems Engineering 2017 – 2021

CONALEP Plantel Mérida I – Technical Baccalaureate in Computer Science 2014 – 2017

LANGUAGES

Spanish: native

English: B2 (TOEFL). Strong technical reading and writing; spoken conversation in progress.